

Audio lecture three. Premature rupture of membranes and preterm prelabor rupture of membranes.

Companion reading handout for simultaneous listening.

This lecture is based mainly on Gabbe, chapter 30, Premature Rupture of the Membranes, Israeli Position Paper number 13 on preterm prelabor rupture of membranes, the local preterm prelabor rupture of membranes protocol, and the AmniSure protocol.

The central tension

Let us begin with the central tension.

P PROM is a race between two dangers. On one side is prematurity. On the other side is infection, cord compression, abruption, and fetal compromise inside a ruptured membrane environment. Management is the art of deciding when time helps the fetus, and when time starts helping the infection.

Premature rupture of membranes, or P R O M, means spontaneous membrane rupture before the onset of labor, regardless of gestational age. Preterm prelabor rupture of membranes, spoken as P PROM, means rupture before labor and before 37 weeks.

Gabbe reports that P R O M complicates about **8 to 10 percent** of pregnancies. P PROM is less common, but it accounts for a substantial fraction of preterm births. Israeli Position Paper number 13 states that P PROM accounts for about **one third of preterm births**, and that in most cases delivery occurs within one week.

That short latency is central. Once membranes rupture, the pregnancy often declares itself quickly. At term, half of expectantly managed patients deliver within **5 hours**, and **95 percent within 28 hours**. Before 34 weeks, Gabbe notes that **93 percent deliver within one week**. Antibiotics can prolong latency, but P PROM is still a condition with a strong clock.

Why membranes rupture

Now let us ask why membranes rupture.

The fetal membranes are not inert plastic. They are living tissue, made of amnion and chorion, with collagen and extracellular matrix. With advancing gestation, physiologic

remodeling weakens the membranes near the internal cervical os. Infection, bleeding, uterine overdistension, smoking, nutritional factors, cervical surgery, contractions, and protease activity can accelerate this weakening.

Infection is especially important. Ascending bacteria can stimulate cytokines, matrix metalloproteinases, prostaglandins, and collagen degradation. This creates a vicious cycle. Infection weakens membranes and promotes rupture. Rupture then makes ascending infection easier.

Diagnosis

Now we move to diagnosis.

The most important diagnostic tool is still clinical. History and sterile speculum examination. If amniotic fluid is seen leaking from the cervix, the diagnosis is made. Pooling in the posterior fornix supports the diagnosis. Ferning and nitrazine can help, but they are imperfect.

Ultrasound can show oligohydramnios, especially if prior fluid was normal, but low fluid alone is not diagnostic. A fetus may have low fluid for renal or placental reasons, and a patient may have ruptured membranes with residual fluid.

Commercial protein tests can help when the diagnosis is uncertain. AmniSure detects placental alpha microglobulin 1, or P A M G 1, in vaginal secretions. The local AmniSure protocol emphasizes that only small amounts of fluid are needed and reports high accuracy, but the P PROM protocol warns that management should not rely on a commercial test alone. This is good clinical discipline. Use the test as part of the clinical picture.

In rare uncertain cases, dye injection into the amniotic cavity, such as indigo carmine, is a gold standard approach, but it is invasive and reserved for special circumstances.

Complications

Now, before we discuss management, let us understand the complications.

Maternal complication

The most common maternal complication after P PROM is chorioamnionitis. Gabbe reports that the risk increases as rupture duration increases, and it is higher at earlier gestational ages. Chorioamnionitis is not just maternal fever. It is an intrauterine inflammatory and infectious process that can injure fetus and mother. It is linked to neonatal sepsis, pneumonia, periventricular leukomalacia, cerebral palsy, and long-term neurodevelopmental injury.

Neonatal complication

The most common serious neonatal complication after P PROM is Respiratory Distress Syndrome, or R D S. This is because P PROM often leads to preterm birth. The earlier the delivery, the greater the burden of R D S, intraventricular hemorrhage, necrotizing enterocolitis, sepsis, and long-term morbidity.

Mechanical risks

There are also mechanical risks. With less fluid, the cord is more vulnerable to compression. Malpresentation increases the risk of cord prolapse. Placental abruption can cause rupture or follow rupture. Fetal distress may result from cord compression, infection, or abruption.

Pulmonary hypoplasia

Now let us focus on pulmonary hypoplasia, because this is one of the most important pathophysiologic ideas.

The fetal lung needs fluid and space to develop. If membranes rupture very early, especially before 20 weeks, and severe oligohydramnios persists for weeks, the developing lungs may not branch and expand normally. Gabbe explains that pulmonary hypoplasia is related to the canalicular phase of lung development. It is not an instantaneous complication. It develops over time after membrane rupture.

The risk is inversely related to gestational age at rupture. It can approach nearly **50 percent** when rupture occurs before 19 weeks with prolonged latency, and it can be extremely high with rupture around 15 to 16 weeks plus persistent oligohydramnios and latency of 28 days. By contrast, lethal pulmonary hypoplasia is rare when rupture occurs after 26 weeks.

The memory hook is: **before viability, P PROM threatens lung construction. After viability, P PROM mainly threatens prematurity and infection.**

Summary before management

Pause here and summarize.

P PROM is diagnosed by history and sterile speculum exam, supported by ultrasound and protein tests when needed. The big maternal complication is chorioamnionitis. The big neonatal complication is R D S. The unique early complication is pulmonary hypoplasia from prolonged severe oligohydramnios.

Management by gestational age

Now we transition to management by gestational age.

Term

At term, management generally moves toward delivery. The fetus is mature enough that the benefit of staying inside is low, while infection risk rises with time.

Near term

Near term, from roughly **34 to 36 weeks**, practice has evolved. Older approaches often delivered after 34 weeks because of infection and cord risks. Israeli Position Paper number 13 notes that expectant management can be considered from 34 weeks to 36 weeks and 6 days in selected cases, because recent data suggest reduced respiratory morbidity without increased neonatal infection. Late preterm corticosteroids can be considered for selected singleton pregnancies without pregestational diabetes and without a prior steroid course, when delivery before 37 weeks is likely.

Remote from term

Remote from term is the classic board-style zone. From **24 weeks to 33 weeks and 6 days**, if there is no infection, no fetal compromise, no significant bleeding, and no other contraindication, expectant management is recommended. The purpose is not to preserve pregnancy forever. The purpose is to gain enough time for corticosteroids, antibiotics, neuroprotection when indicated, and fetal maturation.

Interventions

Let us unpack the interventions.

Antenatal corticosteroids

Antenatal corticosteroids reduce R D S, intraventricular hemorrhage, necrotizing enterocolitis, perinatal death, and long-term neurologic morbidity. Gabbe summarizes meta-analysis data showing that corticosteroids after P PROM reduce R D S, intraventricular hemorrhage, and necrotizing enterocolitis without clearly increasing maternal or neonatal infection.

Antibiotics

Antibiotics are not just infection treatment. In remote-from-term P PROM, they are latency antibiotics. They reduce ascending infection, prolong pregnancy, and reduce neonatal complications. The classic National Institute of Child Health and Human Development, Maternal Fetal Medicine Units regimen used 48 hours of intravenous ampicillin and erythromycin, followed by 5 days of oral amoxicillin and erythromycin. In spoken form, remember it as **2 days I V, then 5 days oral**. Gabbe notes that antibiotics doubled the likelihood of remaining undelivered after 7 days and improved neonatal outcomes. Amoxicillin clavulanate is avoided in this setting because it was associated with increased necrotizing enterocolitis in the O R A C L E data.

Magnesium sulfate

Magnesium sulfate is for fetal neuroprotection when early preterm delivery is anticipated. Gabbe and the local protocol point to use before **32 weeks**. The purpose is not to stop labor. It is to reduce the risk of cerebral palsy and severe motor impairment in very preterm infants.

Tocolysis

Tocolysis is limited. It may briefly prolong pregnancy, but it does not clearly improve outcomes and may increase maternal morbidity. It may be considered selectively to complete steroids in the absence of infection, especially around extreme prematurity, but it is not a default long-term P PROM treatment.

Group B Streptococcus

Group B Streptococcus, or G B S, remains its own issue. Latency antibiotics do not replace intrapartum G B S prophylaxis when indicated.

Monitoring expectant management

Now, how do we monitor expectant management?

We watch the mother, the fetus, and the fluid of the story. Maternal temperature, pulse, uterine tenderness, and symptoms matter. Fetal monitoring matters according to gestational age. White blood cell count can help, but corticosteroids can elevate it, so trends must be interpreted carefully. Ultrasound follows growth, presentation, and amniotic fluid.

If chorioamnionitis is suspected or diagnosed, broad-spectrum antibiotics are started and delivery is pursued. Importantly, intrauterine infection is not itself a contraindication to vaginal delivery. The route of delivery is obstetric.

Cervical cerclage after P PROM

Now let us address cervical cerclage after P PROM.

This is a nuanced area and a likely exam trap.

Gabbe reviews data showing no clear neonatal benefit to retaining the cerclage after P PROM, with trends toward more maternal infection and only brief pregnancy prolongation. Gabbe therefore recommends early cerclage removal when P PROM occurs.

Israeli Position Paper number 13 is more individualized. It states that evidence is inconclusive, so the decision to remove or retain, and the timing of removal, should be based on clinical judgment. However, it clearly says the cerclage should be removed if there is chorioamnionitis or suspicion of active labor.

For exam purposes, if the question is Gabbe-based, early removal is the best answer. For Israeli clinical practice, the answer is individualized, with mandatory removal for infection or labor.

Previaile P PROM

Now let us consider previable P PROM.

Before **23 weeks**, Israeli Position Paper number 13 recommends termination of pregnancy except in cases such as P PROM after amniocentesis, where resealing and better prognosis are possible. If expectant management is chosen, the patient must understand that this is not standard in many contexts and that maternal risks can be substantial.

At **23 weeks to 23 weeks and 6 days**, management is individualized. Counseling must include low survival, long-term disability, maternal infection, abruption, retained placenta, hemorrhage, and the option of pregnancy termination. If expectant management is chosen, antibiotics, betamethasone, and magnesium sulfate may be considered.

This is where communication becomes as important as protocol. You are not just treating membranes. You are guiding a patient through uncertainty where no algorithm can fully predict latency, infection, survival, or disability.

Final synthesis

Now final synthesis.

1. **First, P PROM is a balance between prematurity and infection.** Time is medicine only when it buys maturation without allowing infection or compromise.
2. **Second, diagnose clinically.** Speculum exam is central. P A M G 1 testing can help, but it does not replace judgment.
3. **Third, remember complications.** Chorioamnionitis is the key maternal complication. R D S is the key serious neonatal complication. Early prolonged oligohydramnios risks pulmonary hypoplasia.
4. **Fourth, for 24 to 33 weeks and 6 days, expectant management usually means steroids, latency antibiotics, magnesium before 32 weeks if delivery is imminent, and close surveillance.**
5. **Finally, remember the cerclage nuance.** Gabbe says early removal. The Israeli position paper says individualized, but remove for infection or active labor.

End of lecture three. Pause here before continuing to diabetes in pregnancy.